Step 8 · Evaluation

`src/eval/` · results in `src/eval/results/` · 2026-10-28

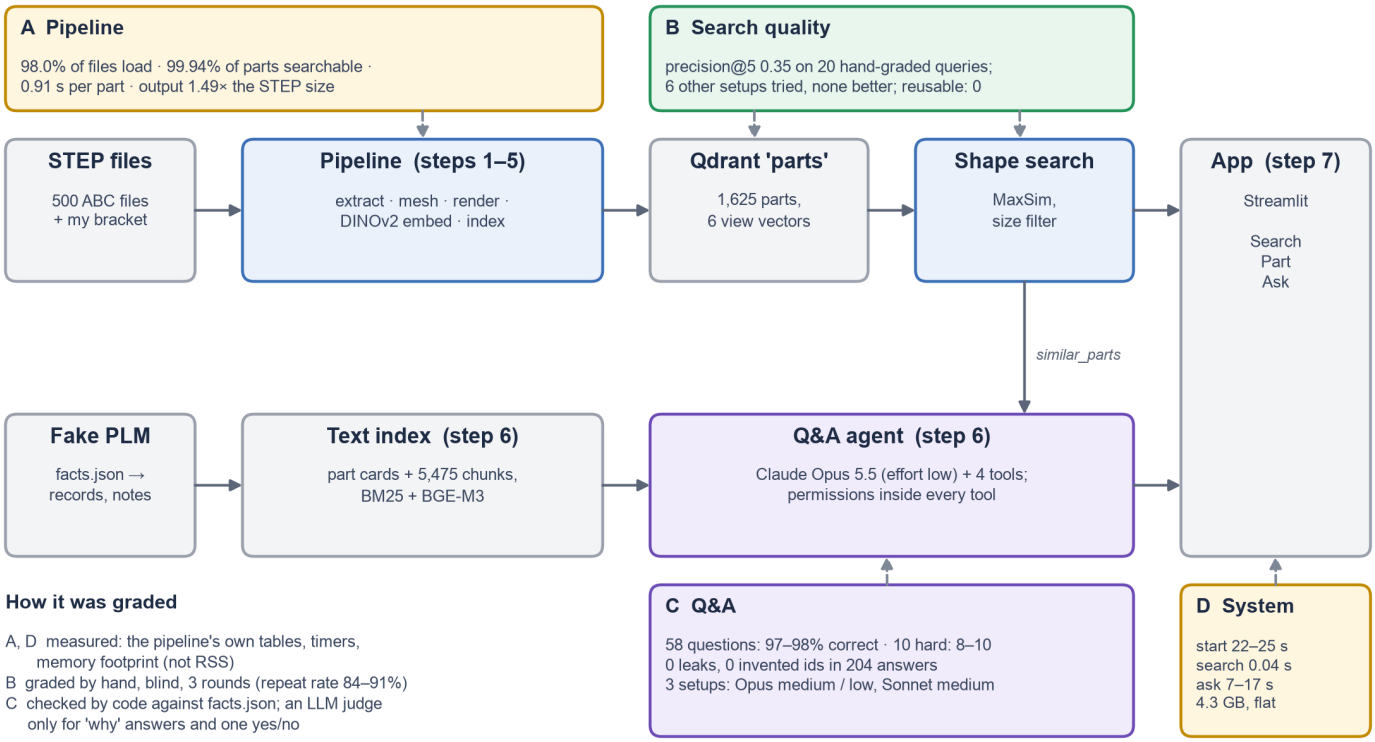


Figure 1. What was measured where, with the headline of each part.

A · Pipeline (500 files, from the pipeline's own tables)

Funnel	Time per searchable part: 0.91 s	Storage: 1.49× the STEP (685 MB)
502 files → 492 loaded (98.0%) 4,613 solids → 1,626 unique (65% in-file copies) 1,626 → 1,625 searchable (99.94%)	embed 41% (0.35 s, steady) · extract 37% (median 0.09 s, max 55 s) · render 13% · mesh 4%	.stl 0.66× (unused by search) · medium .glb 0.39× · Qdrant parts 0.22× · coarse .glb 0.12× · renders 0.10×

D · System (backend timers; app memory as footprint)

Cold start	Search	Part page	STEP attach	Ask (median / p95)	Memory
22–25 s (models 15–17)	0.04 s, cached 0.000	0.04 s	1.1 s, max 5.6	Opus medium 16.9 / 29.8 s; LLM 16.5 s, tools 0.3 s	4.3 GB after load; +66 MB over a 2-tab session; flat over 20 soak rounds

macOS RSS fell 604 → 104 MB under memory compression while the footprint stayed at 4.2 GB: memory is read as footprint. Step 7's '3.15 → 4.2 GB growth' was a reading taken while the models were still loading.

B · Similar-part search (20 queries, 289 pairs graded 0 / 1 / 2 by hand, blind)

Setup	Similar in top 5	vs maxsim per query (win / tie / loss)	Round
maxsim: DINOv2, 6 views, MaxSim (the app)	**0.33–0.35**	—	1, 2
mean: one averaged vector	0.25	2 / 11 / 7	1
maxsim_2x: within 2× size	0.26	3 / 7 / 10	1
aligned: view i to view i only	0.27	1 / 13 / 6 (CI below 0)	1
maxsim_geo: re-ranked by geometry	0.30	3 / 9 / 8	1
voyage: Voyage multimodal-3.5 (API)	0.26	2 / 13 / 5	2
class_maxsim: same PLM class only	0.18	3 / 8 / 9 (CI below 0)	3

- **The real limit is the corpus.** 1,625 unrelated ABC shapes hold almost no part families: no pair of 289 was graded 'could reuse', only 63 distinct similar parts turned up over 20 queries, none for the cover and the pin. A second image model, other scoring and two filters all land near a third: there is little to find.
- **The fix is data:** known variants (e.g. parametric families of the bracket) to score reuse, or, as in production, the full PLM with its part families. Neither fits the demo: time, compute, disk and cost.
- **The app keeps the baseline:** DINOv2-base hosted locally, 6 views, MaxSim. No part image leaves the machine; Voyage scored lower and would send renders to an outside API.
- Grader repeat rate 84–91% on pairs seen twice: differences under ~0.07 are noise.

C · Q&A agent (answers checked by code against facts.json)

	Opus 5.5 medium	Opus 5.5 low (now default)	Sonnet 5.5 medium
58 questions (19 Korean, 5 two-turn)	97%	97%	98%
10 hard (2–4 dependent rounds)	9/10	10/10	8/10
Leaks / invented ids (204 answers)	0 / 0	0 / 0	0 / 0
Seconds, median (main / hard)	9.2 / 18.0	7.5 / 15.7	6.4 / 16.6
$ per question (main / hard)	0.046 / 0.118	0.023 / 0.067	0.018 / 0.045

Opus low over Sonnet: Sonnet's misses were on the multi-step questions; the price gap is ~$0.005–0.02 per question. 10 hard questions decide nothing on their own.

Failures

Where	What broke	Cause → status
Pipeline	9 files no solid, 1 unloadable; 1 part with a 2-triangle mesh	data → dead job / gate 2 rejects; 30 open meshes, 10 invalid B-reps kept and flagged
Pipeline	2 jobs failed with Qdrant down	transient → retried, done
Search	No reusable part found by any of 7 setups	corpus of unrelated shapes, no part families → not fixed in the demo (variants or production data: time, compute, disk, cost)
Search	aligned views worse than MaxSim	canonical pose differs between different parts; MaxSim absorbs it → kept MaxSim
Agent	Greetings ran the id repair (Opus)	copied an example id from a tool description, then made one up → format in words, rule 'never write a made-up id' → fixed
Agent	Blank final answer after 4 tool rounds (Sonnet)	forced last answer came back empty → the last round is announced, the prompt asks for what was found and what is missing → fixed (4/4 honest, forced check)
Tools	Counting change notices by fix gave a lower bound	search returns ≤ 20 → fix_history (all failed tests → change notice → retest) → fixed: both hard questions right in 1 round, Opus low and Sonnet
Eval	First scores 86–88%; 2 hard-set gold answers wrong	scorer and judge too strict, gold checked only the last revision → fixed; models' disagreement found them